

41891: Cloud Computing Infrastructure

42891: Infrastructure for Cloud Computing

Week 4: Cloud Infrastructure Design: Network Virtualization, NaaS

Contents

- * Building IaaS Environment
- * Broadband Networks and Internet Architecture
- * Internet structure: Network of Networks
- * Network Virtualization
 - * VMWare Virtual Networking Concepts
- * Data Center Network Design
 - * Problems: Isolation, Connectivity
 - * Solution: Network Virtualization (Tunnelling)
 - * VXLAN
- * Overlay Network
- * Reference Business Model
- * NaaS

Building IaaS Environment

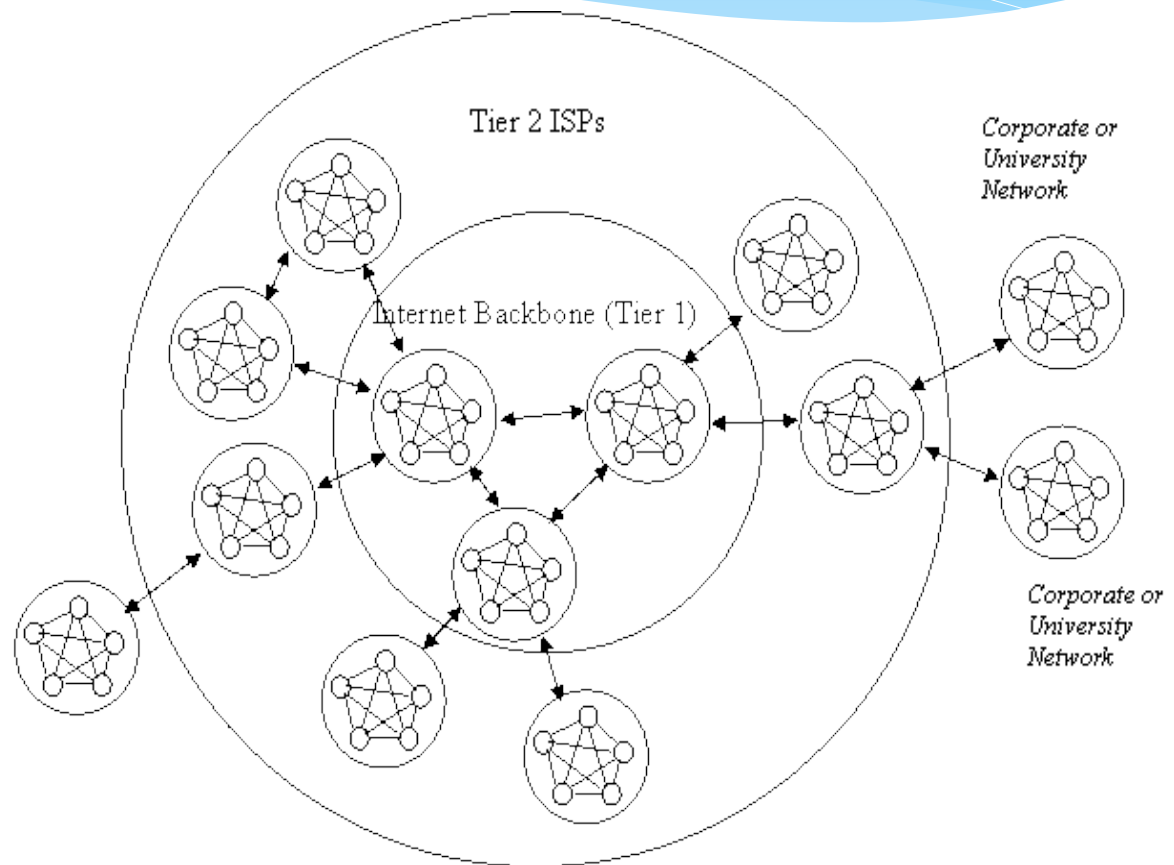
- * Fundamental IT resources that are delivered as part of a standard architecture within IaaS environment: virtual server, cloud storage, and virtual network
- * They are offered in various standardized configurations that are defined by the following properties:
 - * Hardware
 - * Operating software / system
 - * Processors and memory
 - * Servers and / or cloud storage
 - * Virtualized network

Broadband Networks and Internet Architecture

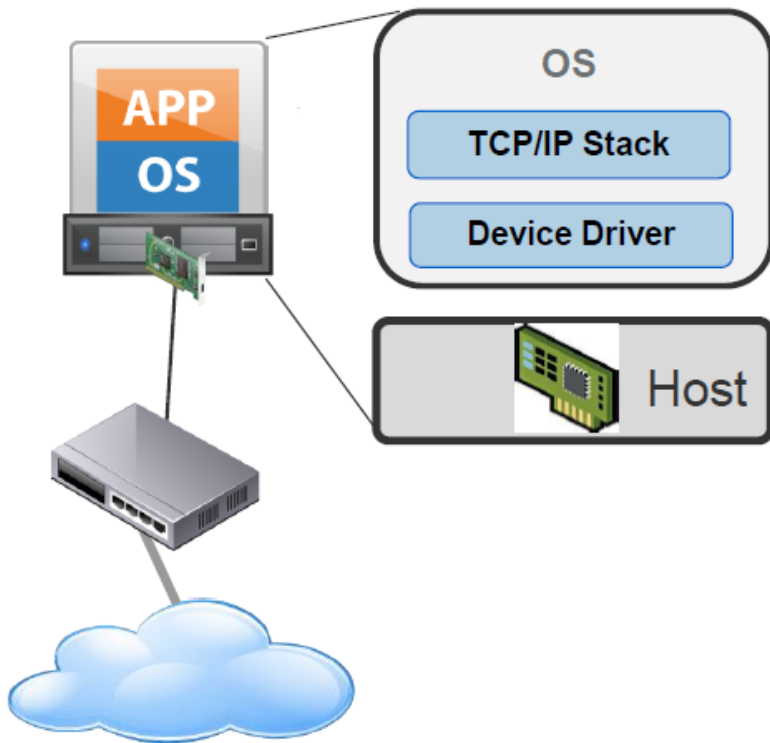
- * All the cloud must be connected to a network
- * Internetworks, or the internet, allows for the remote provisioning of IT resources
- * Internet's largest backbone networks established by ISPs are strategically interconnected by routers that connect the world's multinational networks

Internet structure: Network of Networks

- * **“Tier-1” ISPs** (e.g., Verizon, Sprint, AT&T, Cable and Wireless) known as Internet Backbone networks has international coverage
- * **“Tier-2” ISPs** has regional or national coverage
- * **“Tier-3” ISPs** is the local network of an organization which has hosts connected to it



Physical Networks



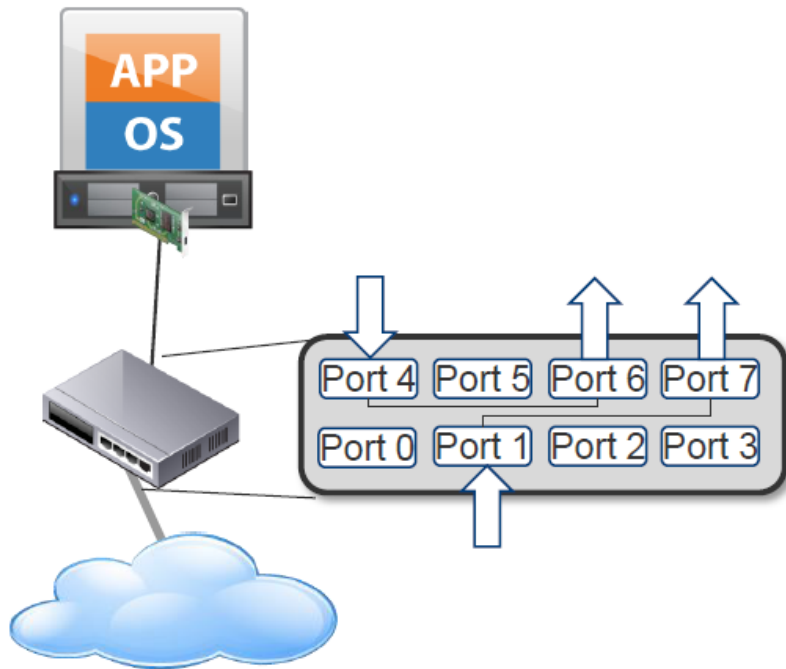
- * Networking on physical host
- * OS runs on the bare-metal hardware
- * TCP/IP Networking stack
- * Network device driver
- * Network interface card (NIC)
- * Ethernet: Unique MAC address

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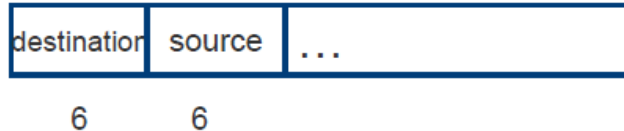
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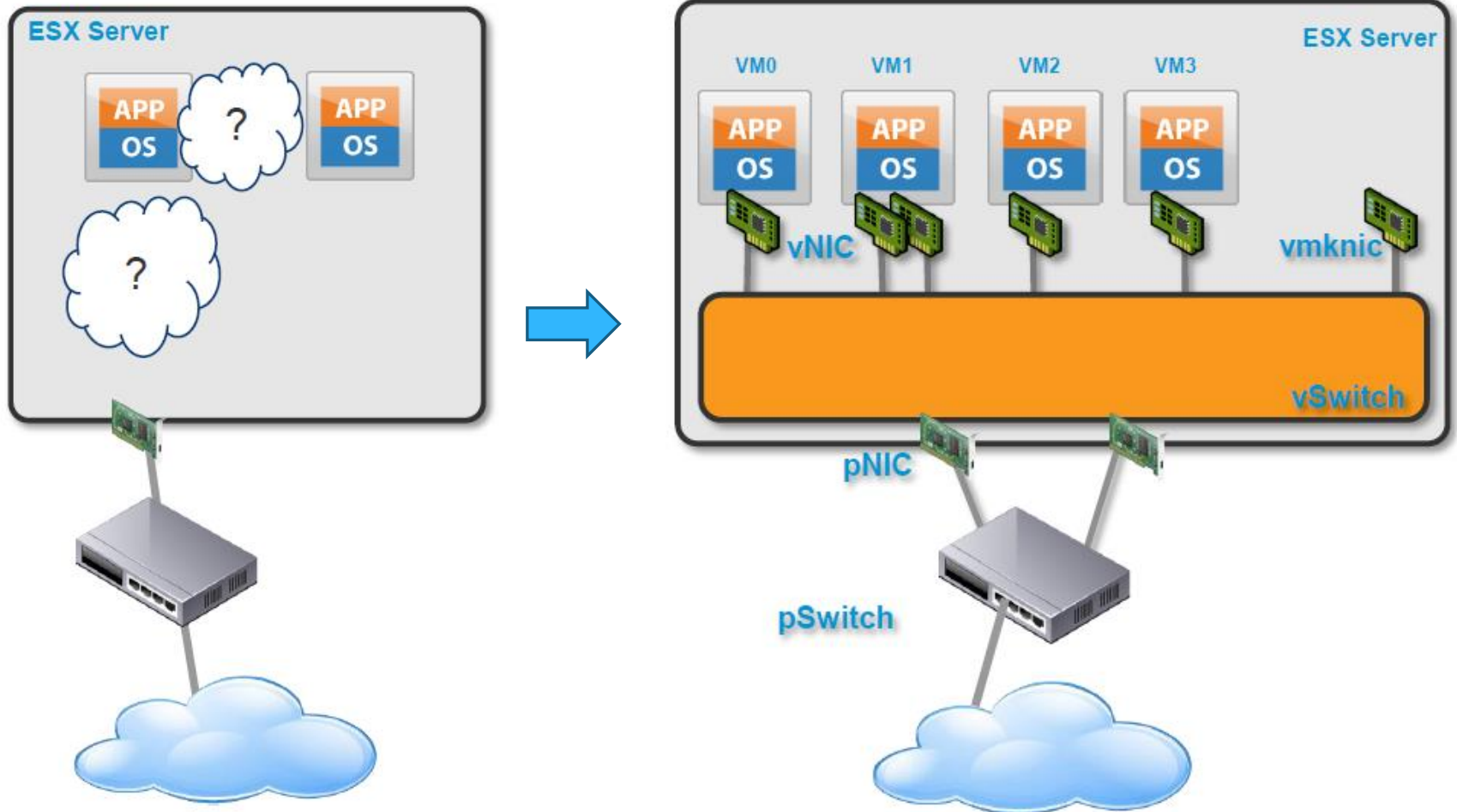


- * Switch is a device that connects multiple network segments
- * Each Ethernet frame contains a destination and a source MAC address
- * When port receives a frame, it reads the frame's destination MAC address

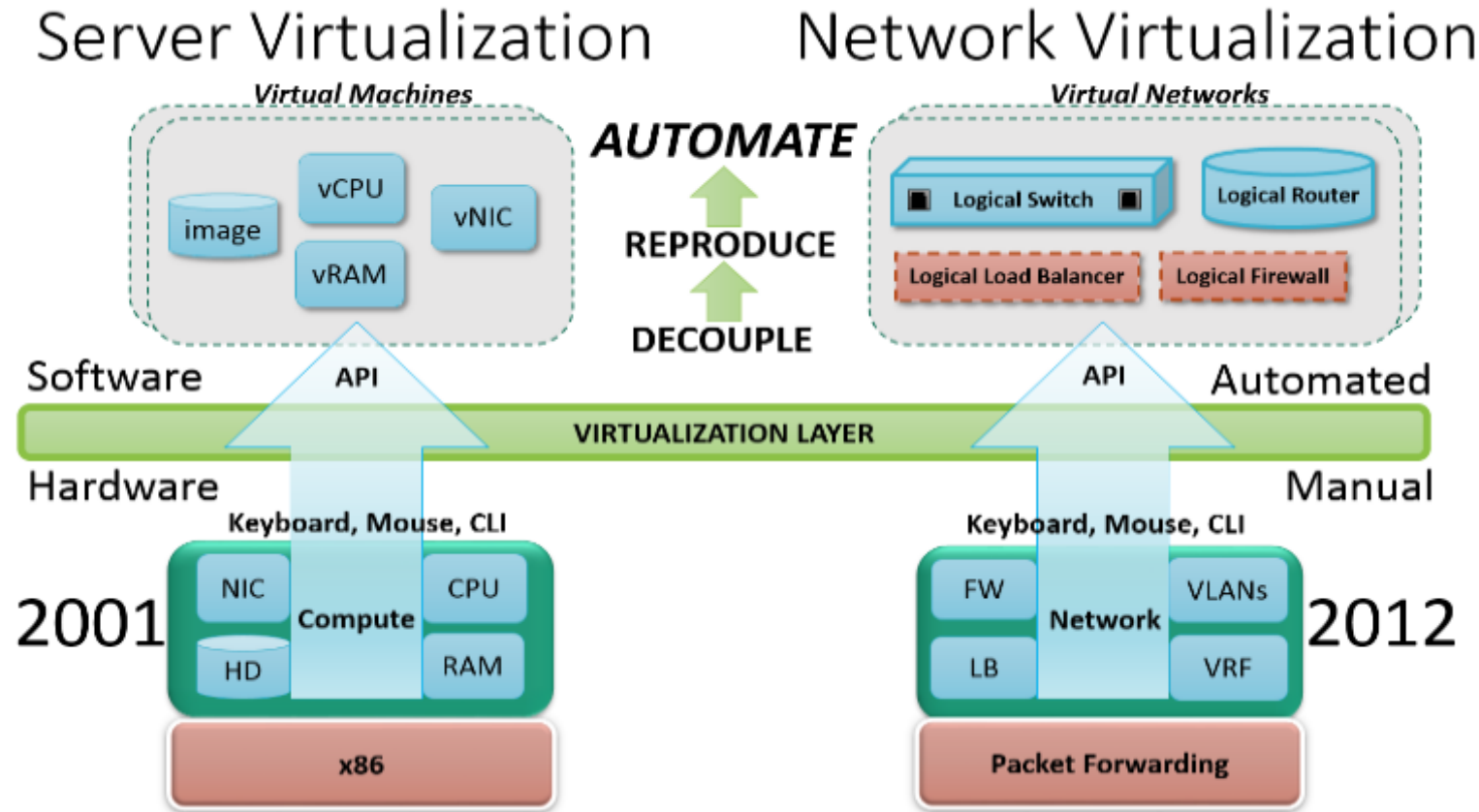
Ethernet frame format



Networking in Virtual Environments



Network Virtualization



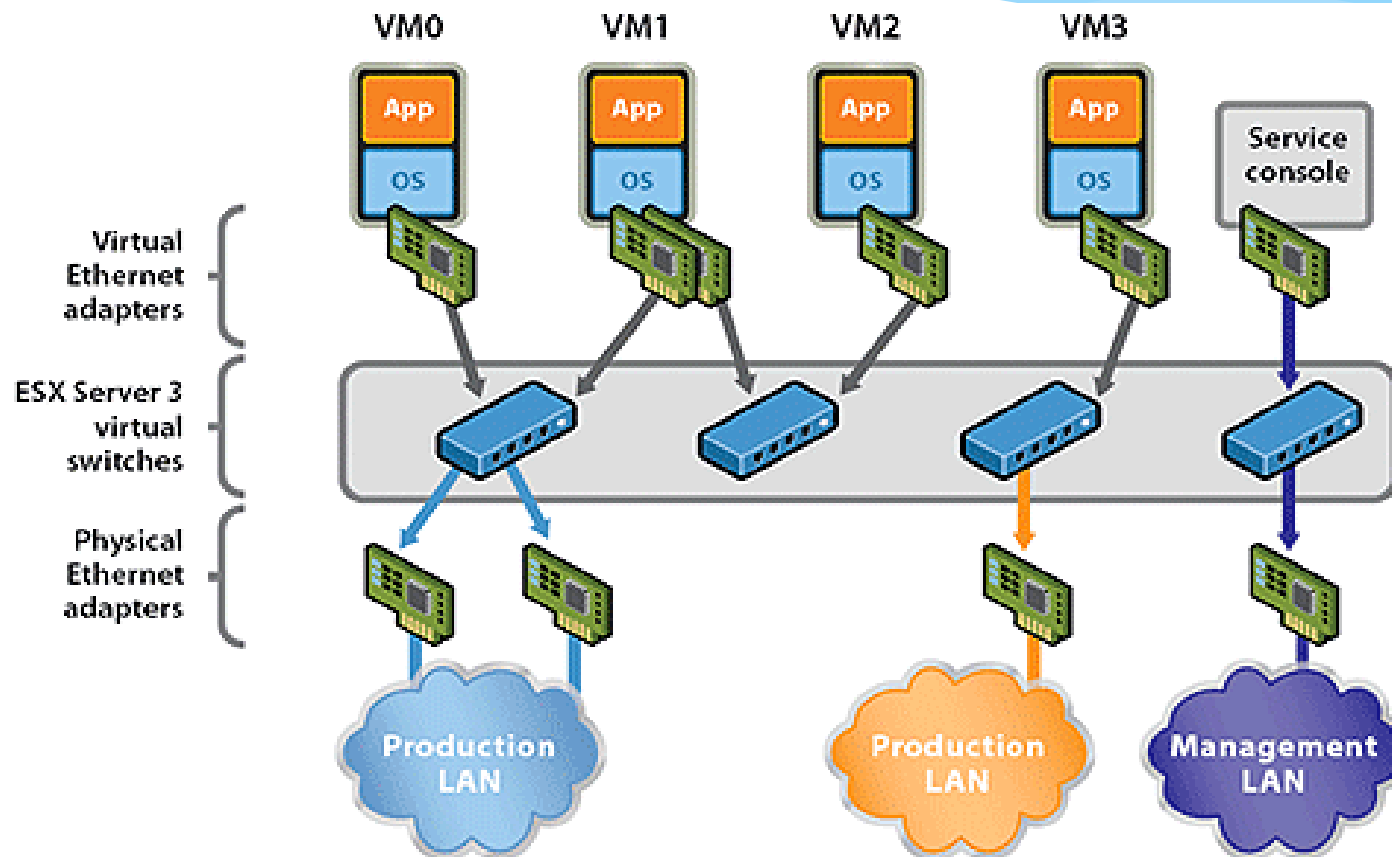
<http://bradhedlund.com/2013/05/28/what-is-network-virtualization/>

<https://www.vmware.com/content/dam/digitalmarketing/vmware/en/pdf/whitepaper/products/nsx/vmware-nsx-network-virtualization-platform-white-paper.pdf>

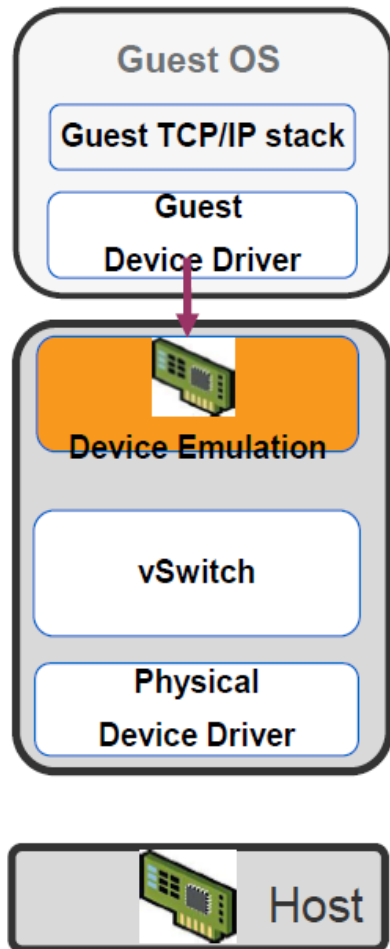
Network Virtualization Environment

- * Allows the creation of multiple virtual networks on same substrate
- * Each virtual network
 - * Is a collection of virtual nodes and virtual links
 - * Is a subset of underlying physical network resources
 - * Co-exists with, but is isolated from, other virtual networks

VMWare Virtual Networking Concepts

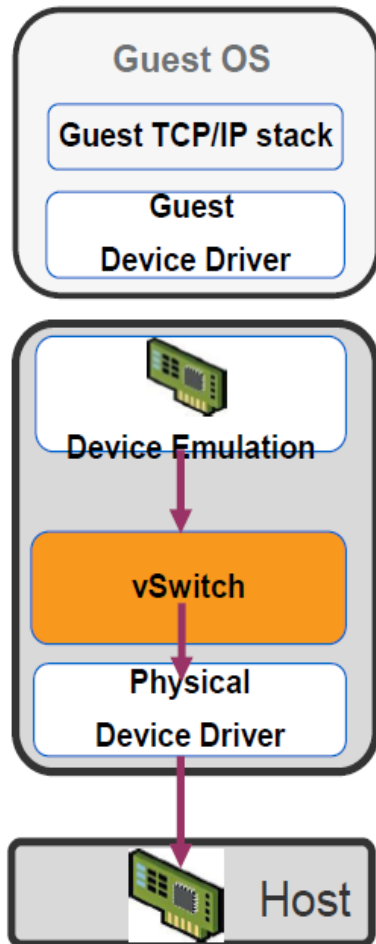


Virtual Network Adapter (vNIC)



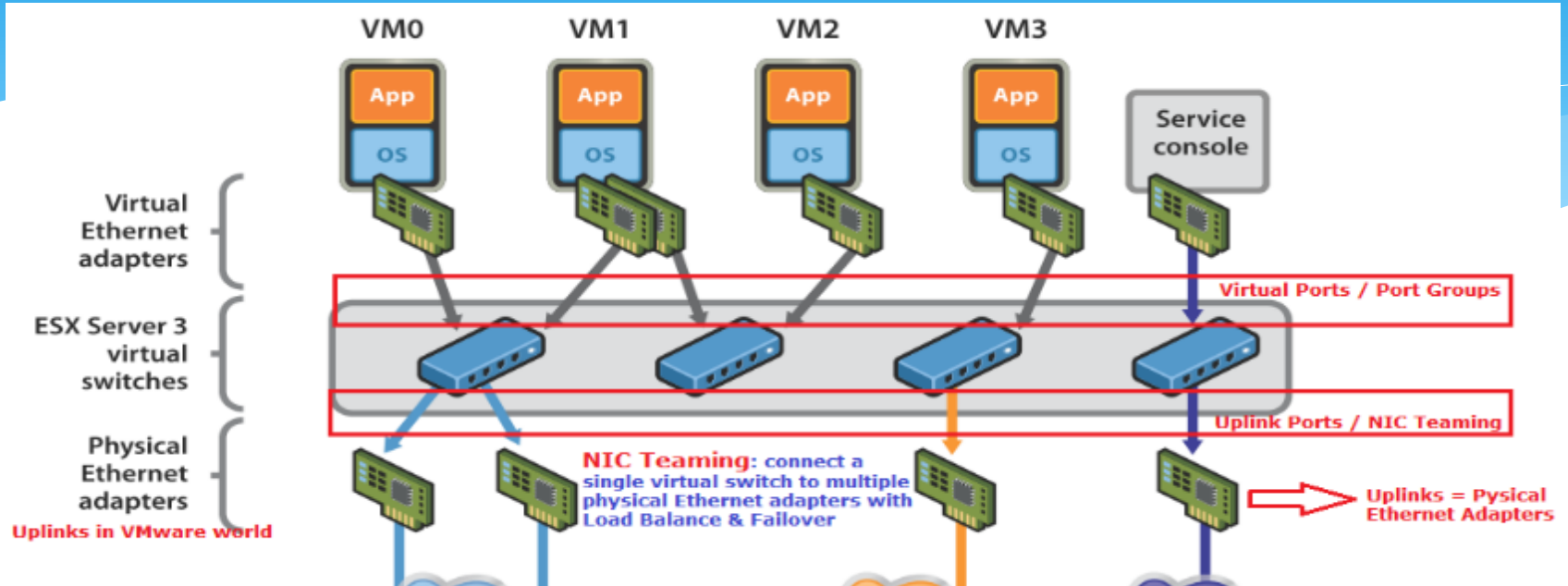
- * What does a virtual NIC implement
 - * Emulate a NIC in software
 - * Implement all functions and resources of a NIC even though there is no real hardware
 - * Each vNIC has a unique MAC address
- * vNICs are completely decoupled from hardware NIC

Virtual Switch (vSwitch)



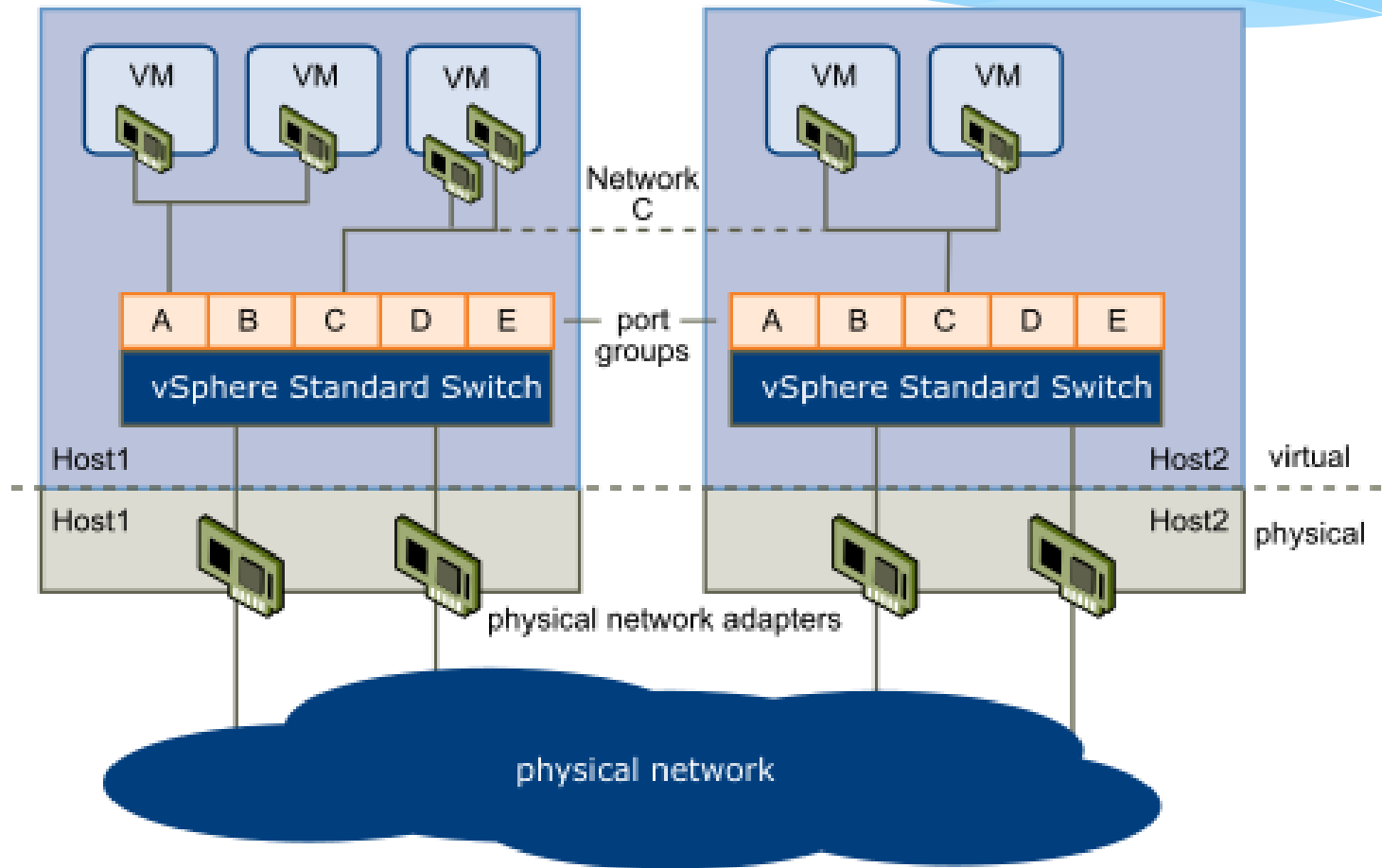
- * How virtual switch works
 - * Software switch implementation
 - * Forwards frames based on destination MAC address
- * vSwitch forwards frames between the vNIC and pNIC
 - * pNIC is shared by all vNICs on same vSwitch
- * Packet can be dispatched to either another VM's port or the uplink pNIC's port
 - * VM-VM
 - * VM-Uplink
- * (optional) bandwidth management, security filters, and uplink NIC teaming

Virtual ports, Uplink ports, Uplinks

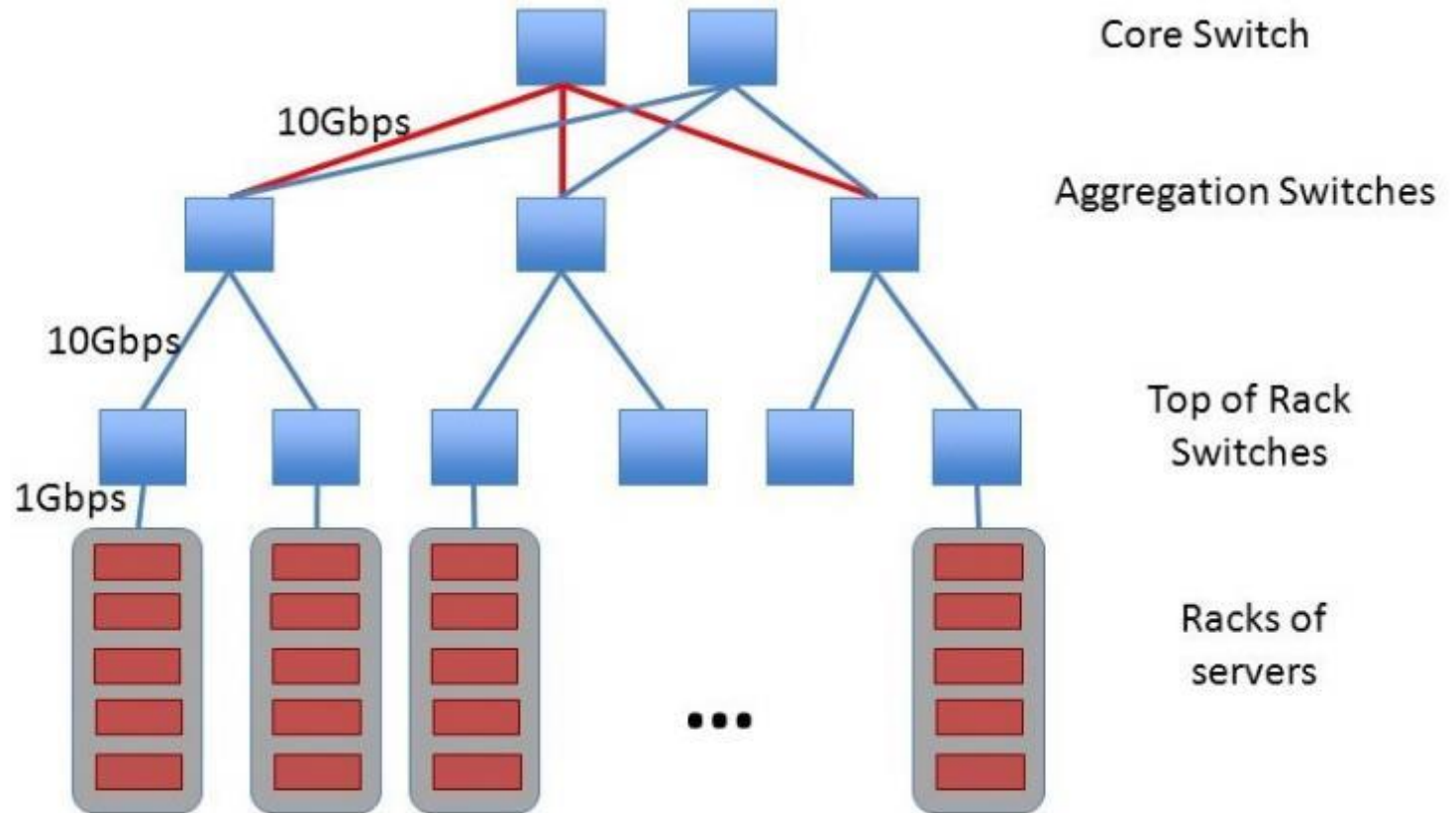


- * **Virtual ports** on a virtual switch provide logical connection points among virtual devices and between virtual and physical devices
- * **Uplink ports** are ports associated with physical adaptors, providing a connection between a virtual network and a physical network
- * **Uplinks:** Physical Ethernet adapters serve as bridges between virtual and physical networks

VMware Network Architecture



Data center network design before VMs



Data Center network design: rack

Front of a rack



Rear of a rack

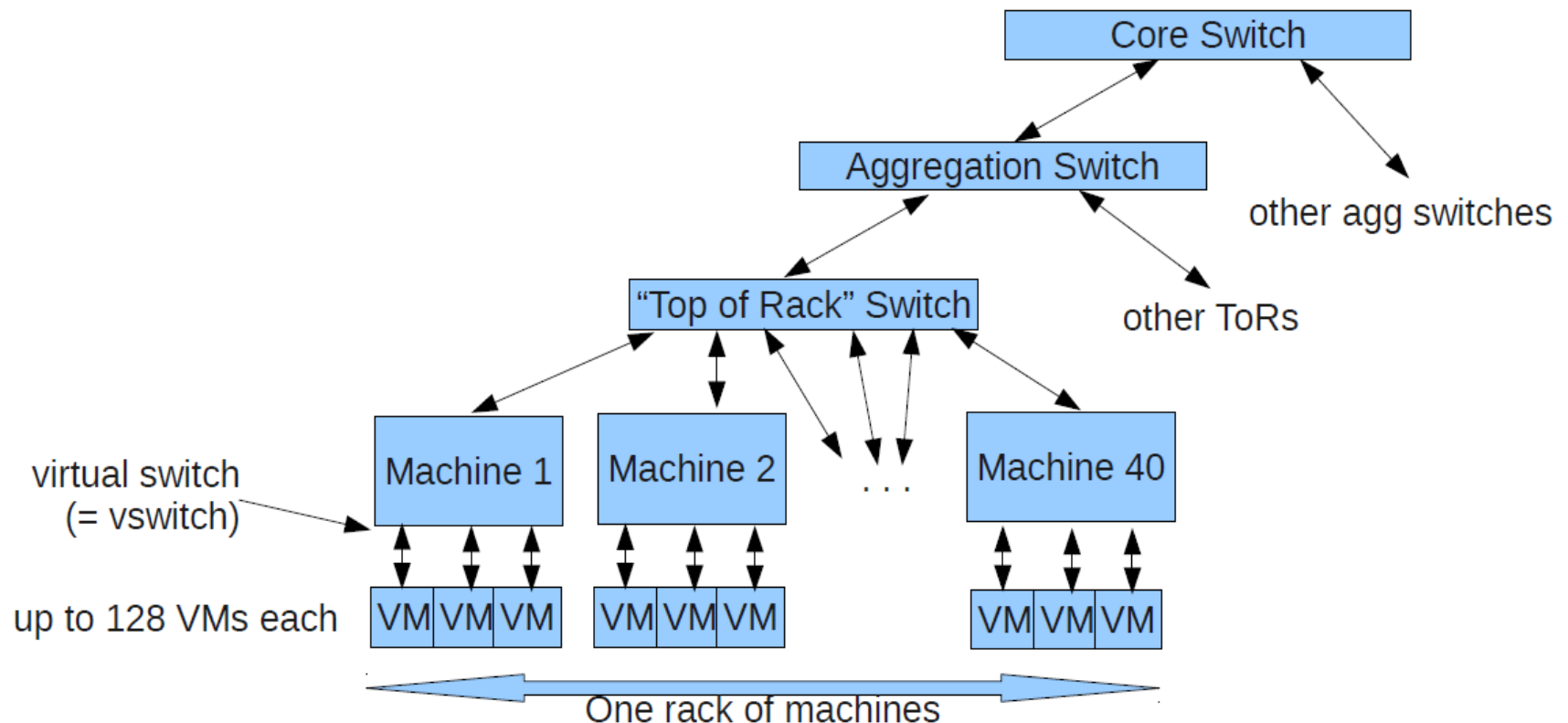


- * Has many racks
- * 1 rack = 20-40 servers
- * Servers in a rack connect to a 48-port “top of rack” (ToR) switch
- * Data center do not rewire their networks without a really good reason

“Top of Rack (ToR)” switch

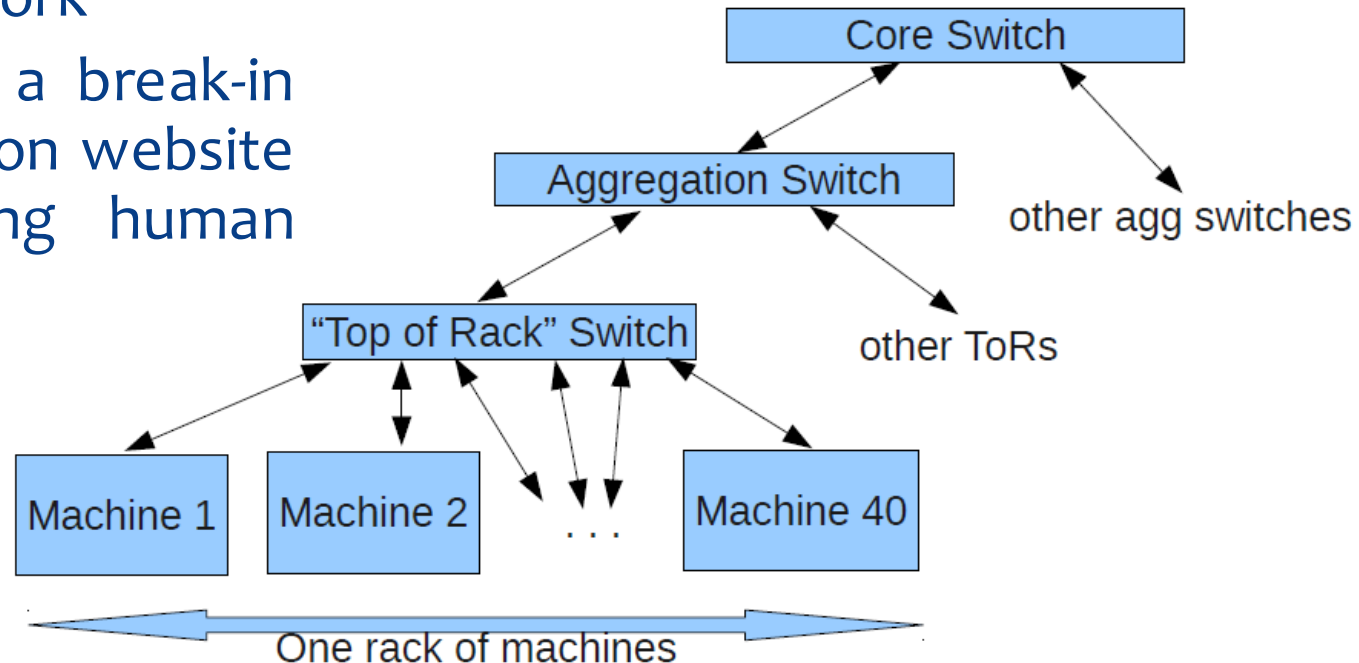


Data center network design with VMs



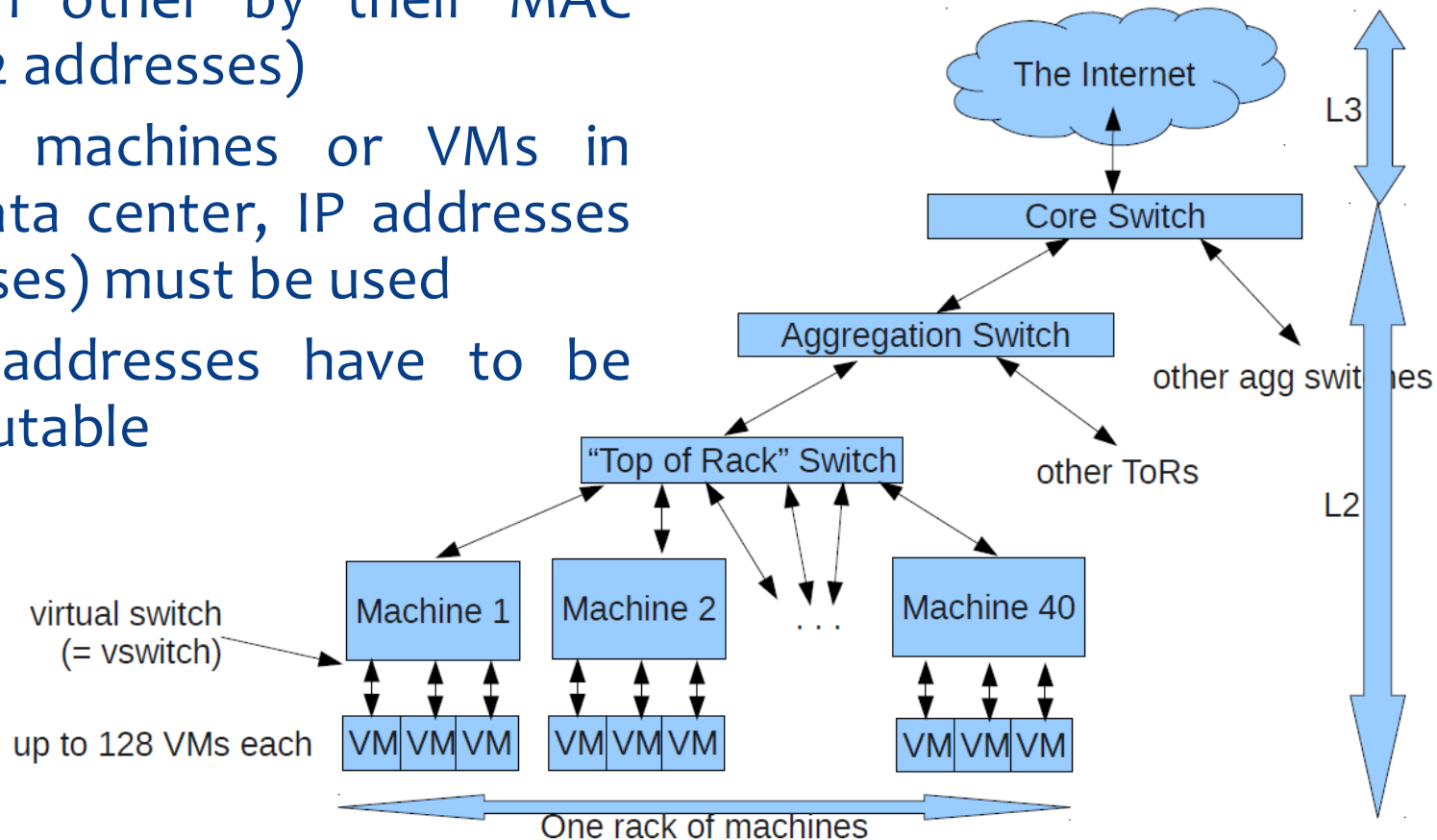
Problem: Isolation

- * All VMs can talk to each other by default
- * You don't want someone in engineering screwing up the finance network
- * You don't want a break-in to your production website to allow stealing human resources data



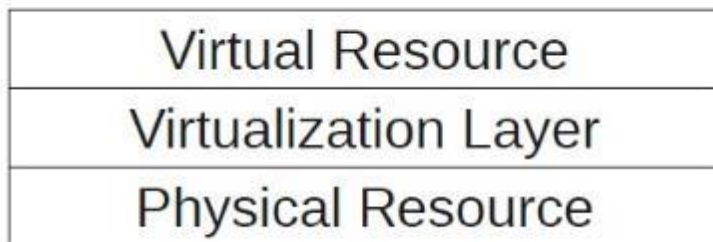
Problem: Connectivity

- * VMs in a same data center can name each other by their MAC address (L2 addresses)
- * To access machines or VMs in another data center, IP addresses (L3 addresses) must be used
- * Those IP addresses have to be globally routable

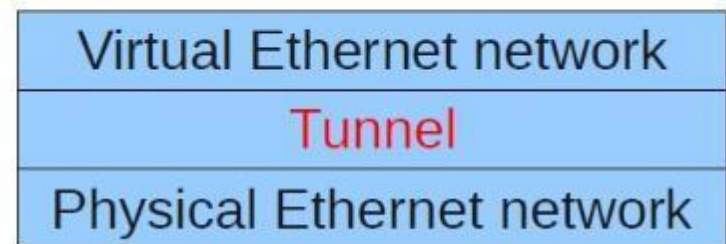


Solution: Network Virtualization

Virtualization Layering



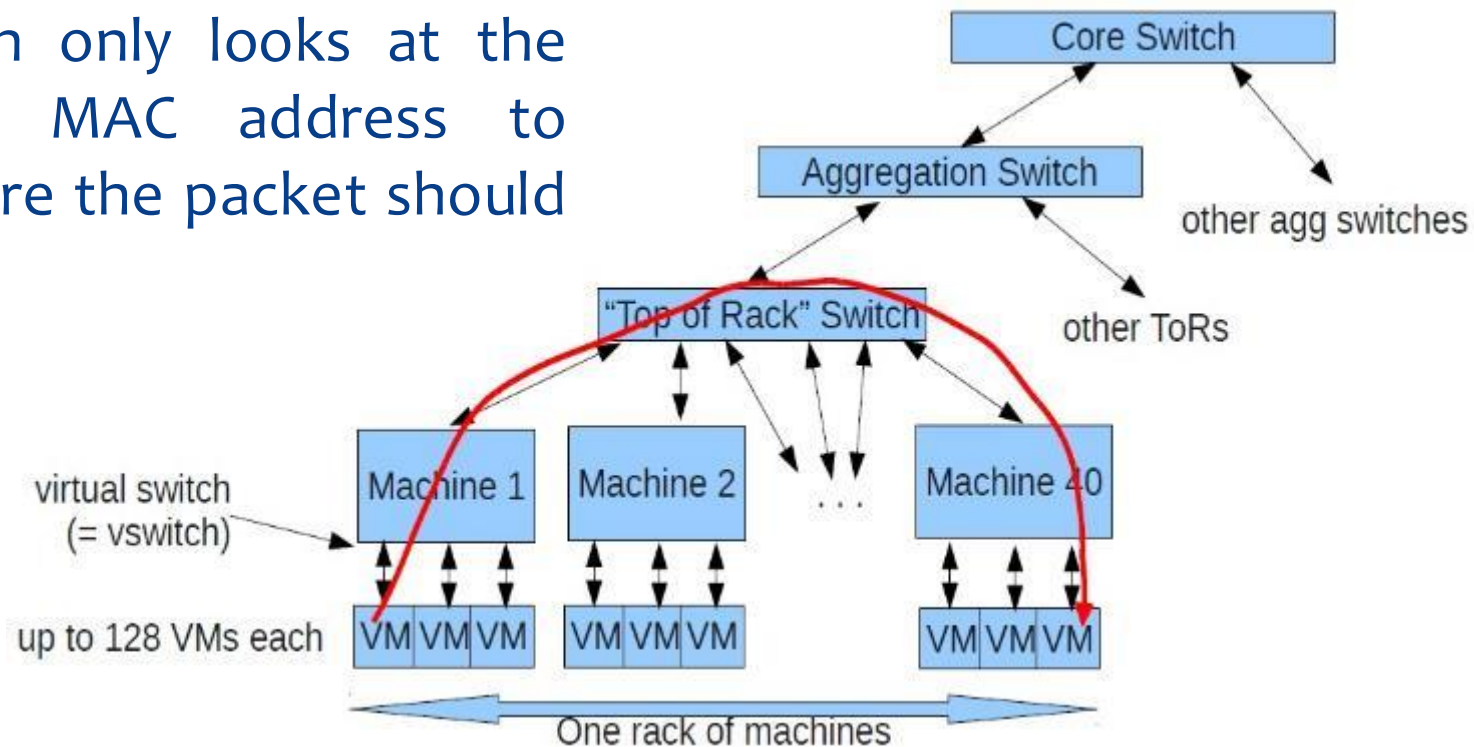
Network Virtualization



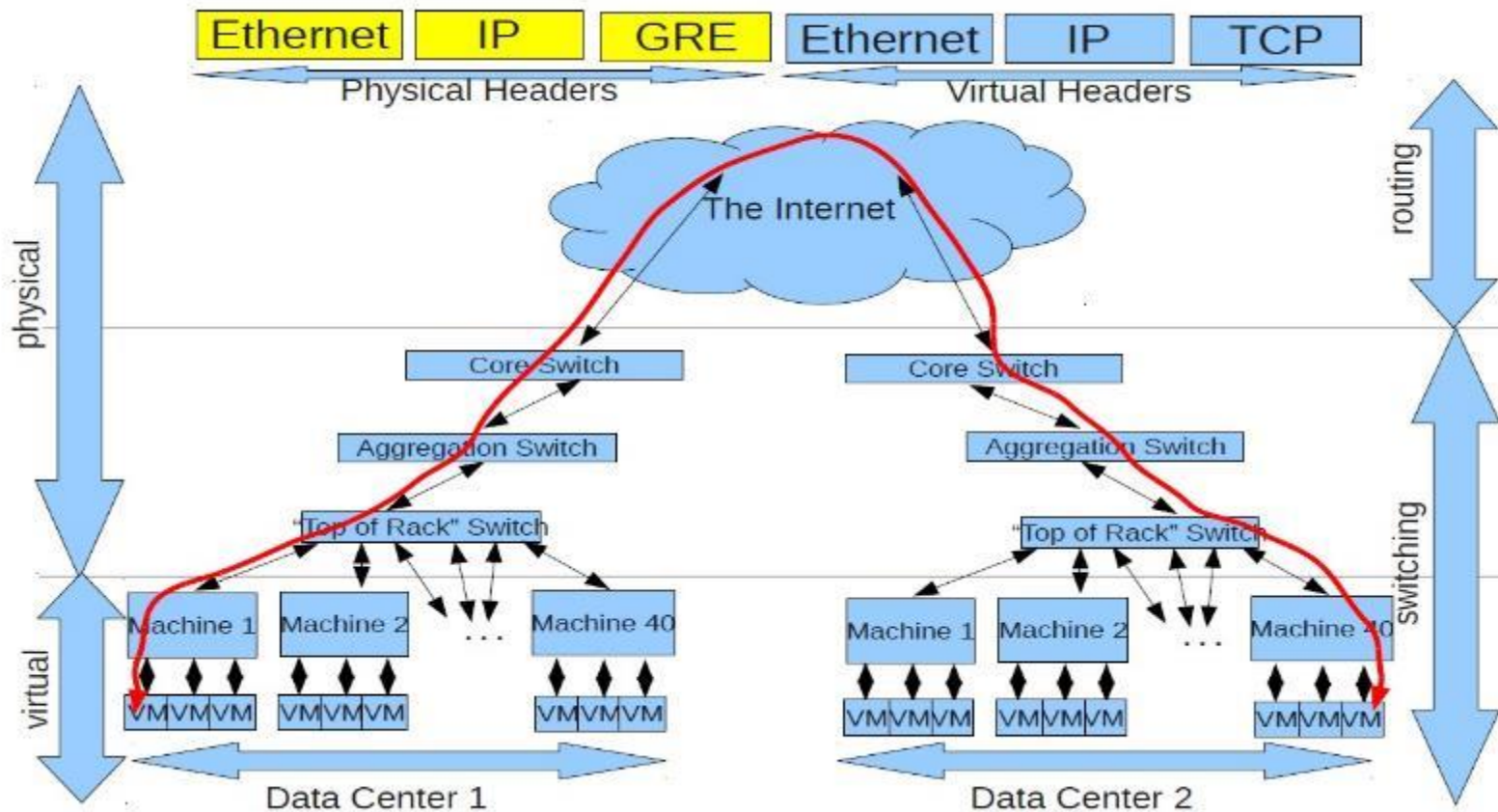
Tunneling: Separating Virtual and Physical Network

Path of a Packet (No Tunnel)

- * Packet from one VM to another passes through a number of switches along the way
- * Each switch only looks at the destination MAC address to decide where the packet should go

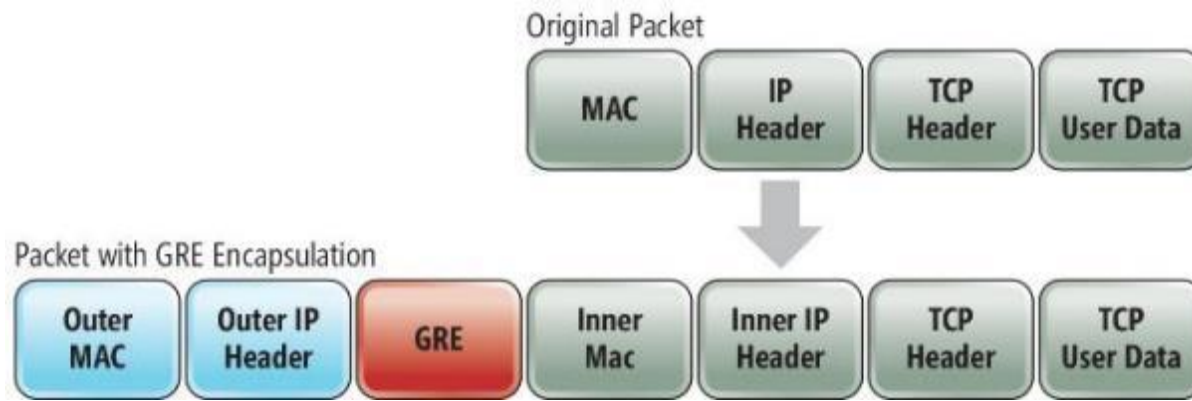


Path of a Packet (via Tunnel)

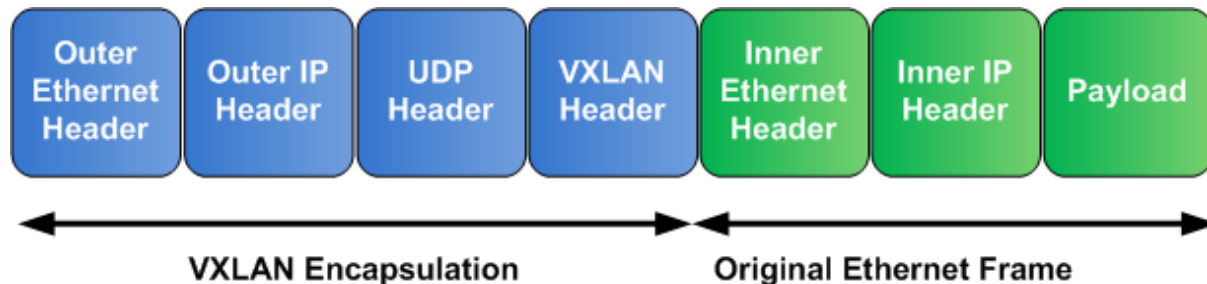


Packet Encapsulation

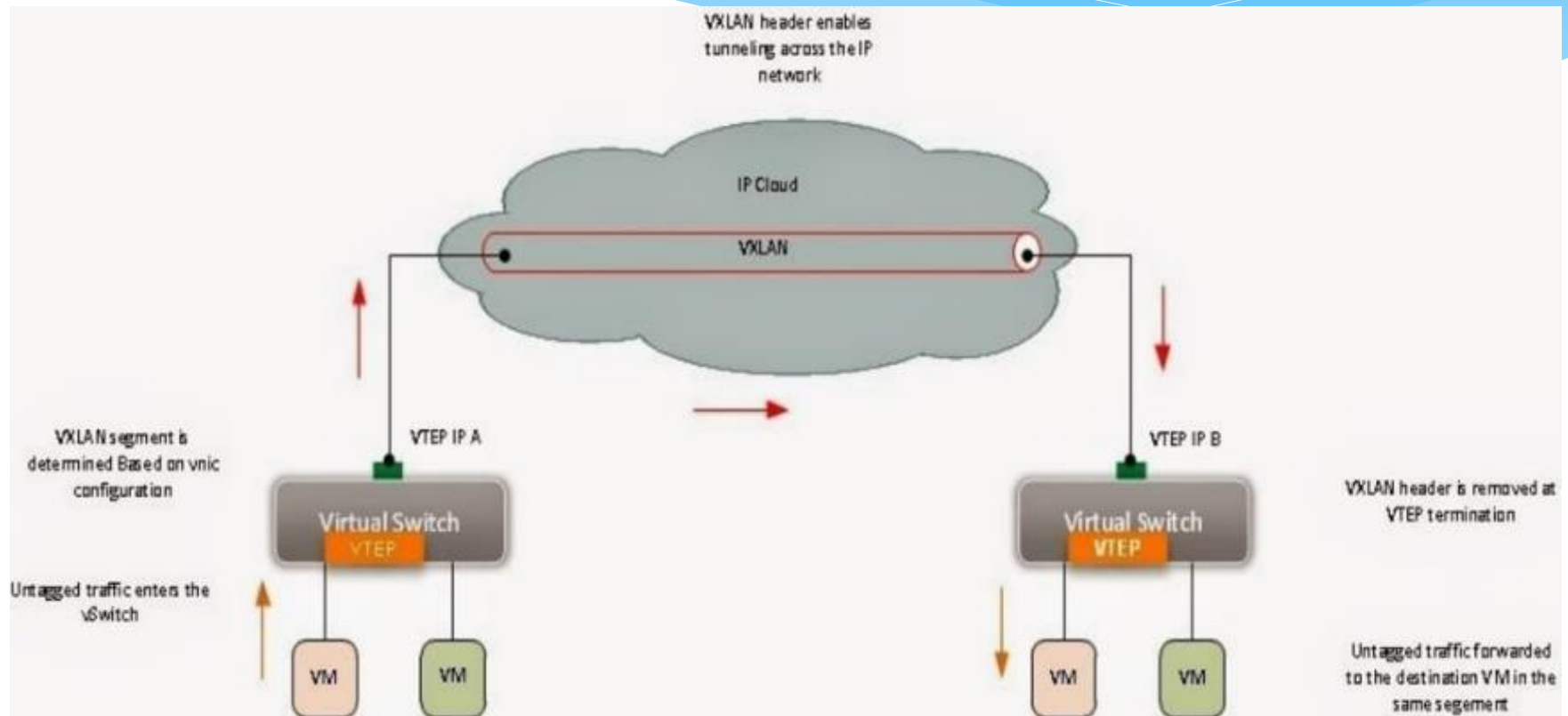
* NVGRE Encapsulation



* VXLAN Encapsulation

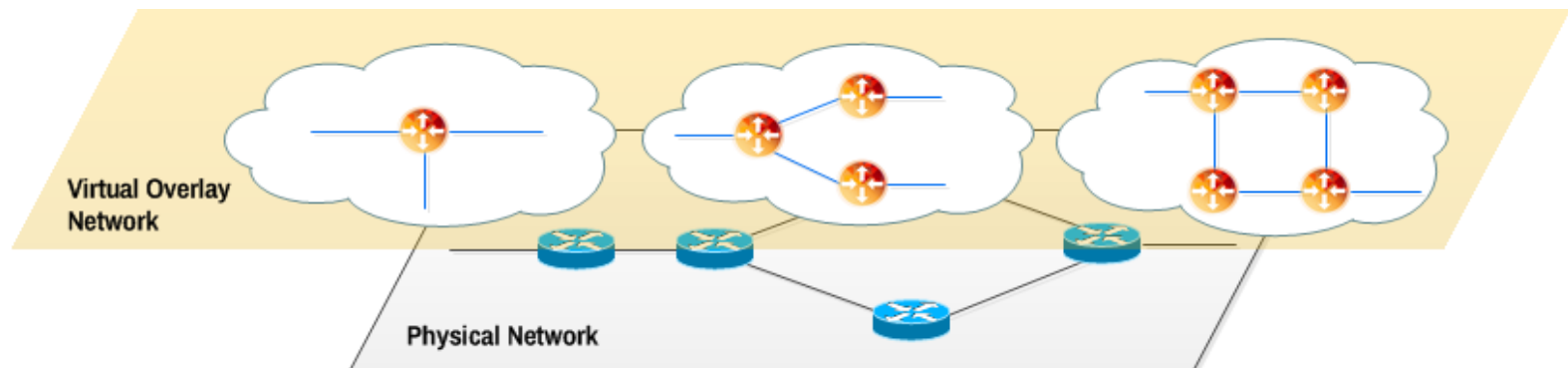


Virtual eXtensible Local Area Network (VXLAN)



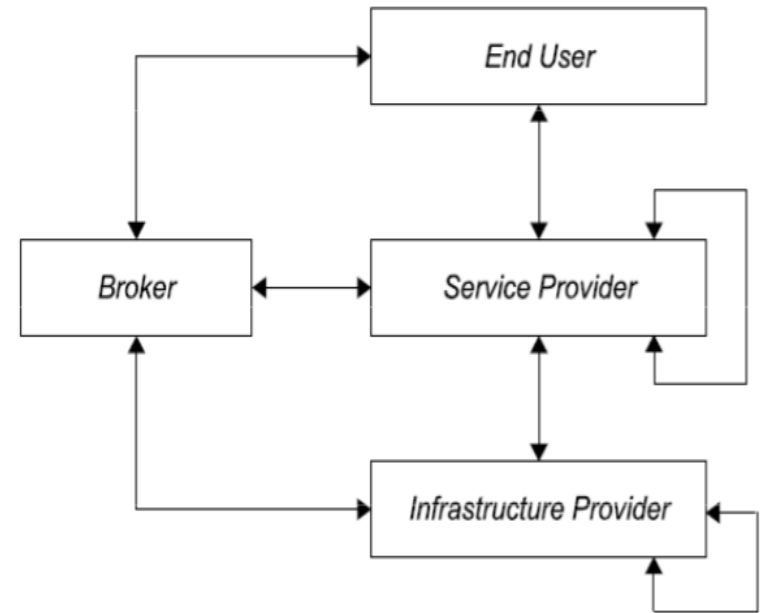
Overlay Network

- * Virtual overlay network is a form of network virtualization that creates virtual layer of network topologies on top of a physical network infrastructure
- * Overlay network protocols include Virtual Extensible LAN (VXLAN), Network Virtualization using Generic Encapsulation (NVGRE), Generic Routing Encapsulation (GRE)



Reference Business Model

- * **Infrastructure providers (InP)**, manages the physical infrastructure
- * **Service providers (SP)**, creates virtual networks by aggregating resources
- * **End users (U)**, who buy and use services from different service providers
- * **Brokers (B)** acts as mediators between InP, SP and U



Relationship between players

Network-as-a-Service (NaaS)

